Topic 3.1 – Climate Change

"We often don't fully appreciate that climate change is a problem. We think it is a problem waiting to happen." – Kofi Annan

Climate change is considered by many to be the planet's greatest threat. We know several of the likely consequences of climate change, most of which we are beginning to experience now. By the middle of the 21st century it is predicted that 200 million people may be permanently displaced due to rising sea levels, floods and drought.

The climate change topic explores variations in the Earth's climate and how both human and natural factors have influenced this. Learners are encouraged to explore why debates around this issue exist before considering its impact on people and the planet.

The future is uncertain and mitigation and adaptation strategies to cope with climate change are evolving. Learners will consider a range of strategies before asking 'can an international response to climate change ever work?'

1. How and why has climate changed in the geological past?

Key Ideas	Content
1.a. The Earth's climate is dynamic.	• Methods used to reconstruct past climate, including marine and lake sediments, ice cores, tree rings and fossils. • Past climate to reveal periods of greenhouse and icehouse Earth, including: ◦ long term, 100 million year transition to colder global climate conditions ◦ glaciation of Antarctica around 35 million years ago ◦ quaternary glaciation ◦ our present interglacial, the Holocene. • How natural forcing has driven climate change in the geological past, including: ◦ plate tectonics, including volcanic activity and continental drift ◦ Milankovitch cycles ◦ solar output ◦ the role of natural atmospheric greenhouse gases.

2. How and why has the era of industrialisation affected global climate?

Key Ideas	Content
2.a. Humans have influenced the climate system, leading to a new epoch, the Anthropocene.	• Evidence the world has warmed since the late-19th century, including: ◦ increases in surface, atmospheric and oceanic temperatures ◦ shrinking of valley glaciers and ice sheets ◦ rising sea level ◦ increasing atmospheric water vapour ◦ decreasing snow cover and sea ice. • Reasons why anthropogenic greenhouse gas emissions have increased since the pre-industrial era. • The balance of anthropogenic emissions around the world and how this has changed in recent history. • How additional greenhouse gases being added to the atmosphere will enhance the natural greenhouse effect. • How humans influence the global mean energy balance. • **Case studies** of **one** AC and **one** EDC to illustrate their contribution to anthropogenic greenhouse gas emissions over time.

3. Why is there a debate over climate change?	
Key Ideas	**Content**
3.a. Debates of climate change are shaped by a variety of agendas.	• How humans have played a part in shaping the climate change debate, including: ◦ historical background of the global warming debate and how it has evolved over time ◦ the role of governments and international organisations, such as the EU or UN ◦ role and possible bias of the media and different interest groups in shaping the public image of climate change.

4. In what ways can humans respond to climate change?	
Key Ideas	**Content**
4.a. An effective human response relies on knowing what the future will hold.	• Overview of climate modelling to illustrate: ◦ importance of the carbon cycle ◦ influence of positive and negative feedback ◦ future emission scenarios, the resulting impacts on global temperatures and sea levels.
4.b. The impacts of climate change are global and dynamic.	• Implications of climate change currently being experienced for people and the environment, such as from changes to ecosystems, health and extreme weather, and how these are projected to change in the future. • The vulnerability of people and the environment to the impacts of climate change.
4.c. Mitigation and adaptation are complementary strategies for reducing and managing the risks of climate change.	• Mitigation strategies to cut global emissions of greenhouse gases, including: ◦ energy efficiency and conservation ◦ fuel shifts and low-carbon energy sources ◦ carbon capture and storage ◦ forestry strategies ◦ geoengineering. • Adaptation strategies to reduce the vulnerability of human populations at risk, including: ◦ framework of adaptation (retreat, accommodate, protect) and its implementation in response to possible future implications of climate change in a range of communities across the development continuum ◦ what future homes, offices, cities, transport and economies will look like following adaptation throughout the 21st century. • **Case studies** of **two** contrasting countries at different stages of economic development to illustrate: ◦ current socio-economic and environmental impacts and the opportunities and threats they present

5. Can an international response to climate change ever work?	
Key Ideas	**Content**
5.a. Effective implementation depends on policies and co-operation at all scales.	• Geopolitics associated with the human response to climate change, including: ◦ role of the Intergovernmental Panel on Climate Change in shaping policy making ◦ success of international directives, such as the Kyoto Protocol ◦ significance of carbon trading and carbon credits ◦ evolution of national, and sub-national policy that extends beyond the vision of international directives.